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STUDENT CASE STUDY EXECUTION REPORT

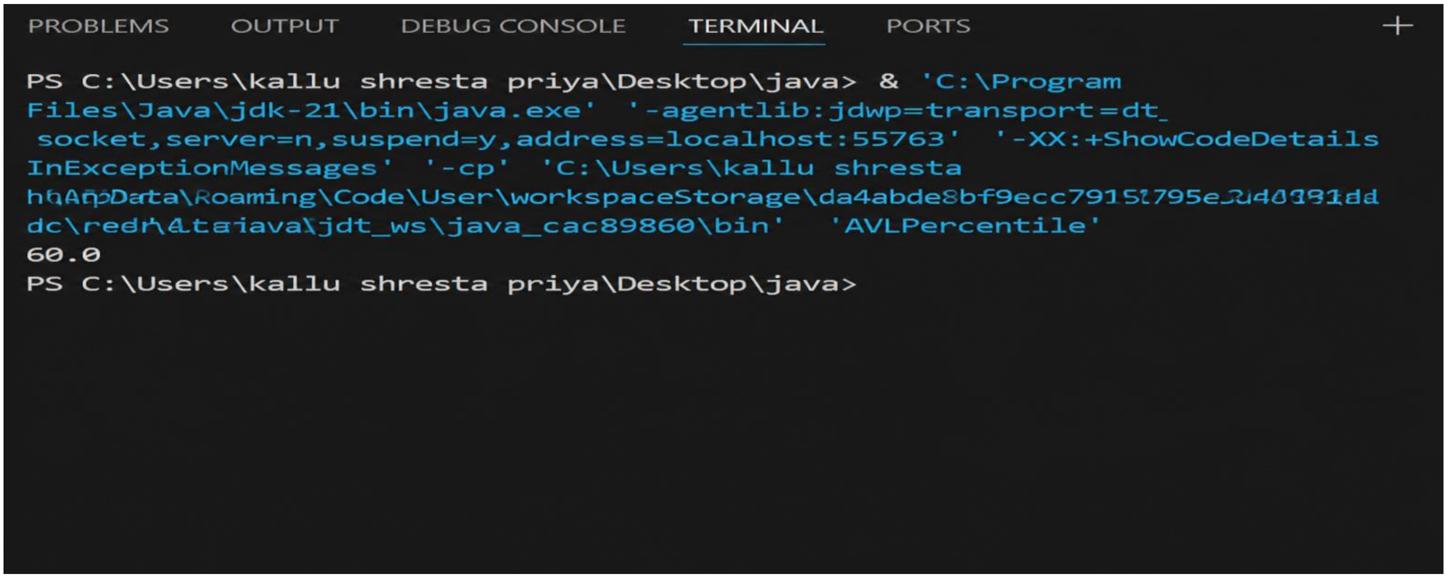
Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	P.Swetha nagasri
Student Name	Kallu shresta priya
Roll Number	2520030127
Branch	CSE
Course Outcome	CO-1

Case Study Topic: Codeforces/LeetCode — Fast Percentile of Submission Computation using AVL Tree with Augmented Count Fields

Work Done Summary:
In this case study, an AVL Tree was implemented to store execution times of accepted submissions in an online judging system. Each node of the AVL Tree was augmented with a subtree size field. The execution times inserted were 120, 85, 200, 65, 150, 95, 180, 75, 110, 240, and 90 milliseconds. Using the subtree size information, the number of submissions less than or equal to a given execution time can be computed efficiently. This enables fast percentile calculation in O(log n) time.

Implementation Details:
1. Created an AVL Tree node containing execution time and subtree size.
2. Inserted all submission execution times into the AVL Tree.
3. Updated subtree sizes after every insertion.
4. Maintained AVL balance using rotations.
5. Stored the size of each subtree at every node.
6. Implemented a function to count submissions less than or equal to a given value.
7. Calculated percentile using the count obtained.
8. Verified logarithmic time complexity for insertion and query operations.

Result: Screenshots / Output



Time Complexity

- AVL Tree Insertion: O(log n)
- Count Query: O(log n)
- Percentile Calculation: O(log n)

GitHub Link: https://github.com/kallushrestapriya-byte/DSA-case-study-1

Student Signature	Faculty Signature